BrightSmile Dental AI Platform

Architecture review, clinical qualification gating, and real-time appointment synchronization.

< 300 ms	99.8%	100%	$34,811
VOICE RESPONSE LATENCY (MODELED TARGET)	INBOUND CALL COVERAGE	DETERMINISTIC RULE GATING	NET ANNUAL CLINIC SAVINGS

End-to-End System Processing Pipeline

INBOUND TO CALENDAR SETTLEMENT

STEP 01 • INBOUND
Patient Engagement
Patients connect through natural voice streamed over WebSocket or interactive chat, communicating concerns in everyday language.

STEP 02 • PERIMETER
Edge Guardrails
Pre-LLM sanitizers screen prompt injection attempts and medical diagnosis queries before token generation begins.

STEP 03 • GATEWAY
Session Broker
Next.js server brokers ephemeral tokens (<60s lifetime). Client audio streams peer-to-peer without exposing core credentials.

STEP 04 • DOMAIN
Clinical Engine
Authoritative TypeScript business rules match patient requests against practitioner credential tiers and operating limits.

STEP 05 • STORAGE
Atomic Commit
PostgreSQL advisory transaction locks coordinate bookings per dentist and date. One selected calendar provider receives each booking; external edits are not locked.

Practitioner Credential Tiers & Clinical Scope

HARD BOUNDARY RULE MATRIX

CLINICAL PROCEDURE	DURATION	DR. JUNIOR (GENERAL)	DR. SENIOR 1 & 2 (SPECIALIST)
Routine Teeth Cleaning	30 min	Permitted Standard Care	Permitted Specialist Overseen
Laser Teeth Whitening	45 min	Permitted Cosmetic	Permitted Full Scope
Root Canal Treatment	90 min	Restricted Endodontic	Permitted Primary Specialist
Wisdom Tooth Extraction	60 min	Restricted Surgical	Permitted Oral Surgery Lead
Full Mouth Reconstruction	120 min	Restricted Surgical	Permitted Surgical Lead

Deterministic Scope Enforcement
Medical authorizations are enforced deterministically in the application core (`AppointmentService`), never left to the discretion of an LLM prompt. If a patient attempts to schedule oral surgery with Dr. Junior, the scheduling engine rejects the assignment and seamlessly reroutes to an available senior doctor.

Empathetic Rerouting Dialogue
"Dr. Junior specializes in routine hygiene and whitening. For root canal therapy, our lead endodontist Dr. Senior 1 has openings this Thursday at 2:00 PM. Shall I reserve that for you?"

Operational Safety & Zero-Trust Governance

PATIENT PROTECTION STANDARDS

Adversarial Prompt Shield
Pre-parsing filters reject prompt injection patterns (such as instructions to ignore clinical rules or extract internal instructions) before triggering downstream LLM context, ensuring predictable behavior and avoiding wasted compute.

Clinical Advice Deflection
The conversational assistant strictly avoids providing diagnoses or medication prescriptions. Any inquiry regarding acute distress triggers clear guidance directing the patient to seek urgent clinical evaluation.

Privacy & Credential Hygiene
Primary API secrets and OAuth credentials stay strictly on the server. Audio is processed transiently in memory, preventing persistent capture of voice recordings while maintaining patient confidentiality.

Voice Architecture, Calendar Sync & Financials

Conversational voice mechanics, selectable calendar providers, and illustrative practice ROI modeling.

Realtime Voice Infrastructure

STREAMING WEBSOCKET VOICE • PROVIDER-AGNOSTIC

Latency & Interruption Profile

Traditional Chained Pipeline (STT → LLM → TTS) ~2,500 ms

Noticeable turn pauses; difficulty handling natural mid-sentence interruptions.

BrightSmile Direct Streaming Voice (PCM over WebSocket) < 300 ms

Full-duplex audio; responsive interruption detection feels fluid and human.

Three-Phase Lifecycle Execution

1. Session Handshake: The browser asks `GET /api/v1/voice/session` which provider is active, then `POST` s for the session descriptor. With Gemini Live the backend mints a single-use token, keeping long-term keys secure on the server; an OpenAI Realtime-compatible endpoint needs no token.

2. Direct Media Stream: Audio streams as PCM16 over WebSocket — 16 kHz up, 24 kHz down — without a separate speech-to-text hop.

3. Server-Verified Tool Calling: When an appointment is scheduled, function calls are relayed to `/api/v1/voice/tools` for authoritative database validation.

Calendar Synchronization & Atomic Integrity

PROVIDER PARITY & CONFLICT PREVENTION

Google Calendar Adapter

Executes batch FreeBusy interval queries for target practitioner calendars. Manages proactive OAuth 2.0 token refreshes in memory 5 minutes prior to expiry.

Microsoft 365 Outlook Adapter

Connects to Microsoft Graph `calendarView` via Entra ID app credentials. Recursively drains paginated busy blocks and normalizes timestamps to UTC RFC3339 format.

Dynamic Database Settings

Practitioner calendar addresses are stored in a dedicated database table, enabling clinic administrators to adjust calendar mappings through the web UI without redeploying code.

ATOMIC CONCURRENCY MODEL: Postgres advisory transaction lock per dentist and date → check local bookings and both calendars' availability → create booking

Operating Economics & Practice Staffing ROI

TRANSPARENT COST ANALYSIS

INFRASTRUCTURE SERVICE	MONTHLY WORKLOAD	MONTHLY COST	SHARE
Realtime Voice Audio (illustrative: OpenAI Realtime rates as a stand-in)	3,000 Spoken Minutes	**$555.00**	92.6%
OpenAI Chat Assistant (GPT-4o-mini)	15,000 Conversation Turns	**$4.05**	0.7%
Next.js Application Hosting	150,000 HTTP Requests	**$20.00**	3.3%
Managed PostgreSQL Database	10 GB SSD Storage	**$15.00**	2.5%
Google & Microsoft Cloud APIs	FreeBusy Schedule Inquiries	**$5.00**	0.8%
Total Operating Expense	**~650 Patient Appointments**	**$599.05**	**100%**

Traditional Receptionist $3,500 / mo

Covers standard business hours; missed evening calls and peak phone lines create lost booking opportunities.

BrightSmile AI Engine $599.05 / mo

Browser voice and chat depend on application and provider availability; telephony and guaranteed response times are not included.

ANNUAL PRACTICE SAVINGS
$34,811 / yr
83% Net Cost Reduction

VERIFICATION STATUS: Automated tests use mocks for external providers. Live voice and calendar integration require separate validation. Validation Required